

George Petrakis

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EDUCATION

University of Ioannina

BSc in Informatics and Telecommunications

- Focused on Software Engineering, Network Architectures, and Cyber Security.

Ioannina, Greece

Sep 2022 – Present

University of Ioannina

Certificate of Pedagogical and Teaching Competence (Grade: 9.8/10)

- Qualified to teach Computer Science in Secondary Education. Conducted an internship at a General High School in Arta, instructing the "GLOSSA" programming language to 11th-grade students.

Ioannina, Greece

Feb 2024 – Jun 2026

5th EPAL of Heraklion

IT and Networks Technician (GPA: 18.8/20)

Heraklion, Crete, Greece

Sep 2019 – Jun 2022

EXPERIENCE

Cybersecurity Engineering Intern

Laboratory of Knowledge and Intelligent Computing KIC (EPIC) - UoI

- Access Control System:** Developed a secure Client-Server authentication and age verification system for Metaverse environments (Roblox).
- Backend & Security:** Engineered a robust backend using Node.js, Express, and MongoDB, integrating critical security mechanisms such as NoSQL Injection protection, XSS mitigation, and Rate Limiting.
- Identity Management:** Implemented the eIDAS Trust Triangle model to ensure the secure issuance of digital credentials and single-use tokens.
- API Integration:** Successfully connected the external authentication system with the virtual game world utilizing secure REST APIs.

Mar 2026 – Apr 2026

Arta, Greece

PUBLICATIONS

A Local-First Event-Driven Order Management System for Temporary Events – Submitted to IEEE SEEDA 2026 conference.

Co-authored with S. Margariti & C. Gogos.

Security architecture and the protection of minors in the Metaverse – Co-authored during the KIC EPIC Internship (In Preparation / Under Review).

PROJECTS

UmbrellaAgent (Bachelor Thesis) | MERN Stack, Socket.IO, Hardware Printing

- Developed a local-first, WebSocket-driven Order Management System for high-density environments.
- Engineered a custom ESC/POS thermal printing service over raw TCP with Greek encoding support.
- Validated system resilience via Chaos Engineering achieving an RTO under 5s, and sustained over 1.58M requests in Apache JMeter stress tests.

Arduino WSN ESPNow | C++, ESP32, ESP-Now Protocol

- Developed a low-latency Wireless Sensor Network (WSN) using ESP32 and ESP-NOW protocol for power-efficient data transmission.
- Implemented embedded routines involving GPIO control, hardware interrupts, and state machine logic in C++.

Social Network Platform | React, Node.js, Express, MongoDB

- Engineered a scalable full-stack social media application, utilizing React for a dynamic frontend and Node.js/Express for robust backend architecture.
- Designed RESTful APIs for user interactions and managed persistent, NoSQL data storage with MongoDB, ensuring secure authentication and authorization.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C/C++, Python, SQL, Lua, VHDL, PHP | **Databases:** MongoDB, SQLite

Web Technologies: React.js, Node.js, Express.js, WebSockets, Tailwind CSS

Tools & Platforms: Git, Docker, Linux, Roblox Studio, Arduino IDE, Wireshark, Packet Tracer

CERTIFICATIONS

Develop AI-generated Prototypes in Google AI Studio – Google Cloud Skills Boost (May 2026)

NDG Linux Unhatched & Intro to Cybersecurity – Cisco Networking Academy (2023)

Proficiency (C2) in English – LRN (2022)